Qalb E Zahra Mashhadi's Score

88%

✦ Great work!

15 correct out of 17 questions

15	2	0
Correct	Wrong	Skipped

FULL ANSWER KEY WITH EXPLANATIONS

Q1 SCENARIO Identify TP & FP ✓ Correct

TP = 40, FP = 20

Of the 60 predicted positives, 40 are truly diabetic → TP = 40. The remaining 60 − 40 = 20 were healthy patients wrongly flagged → FP = 20.

Q2 SCENARIO Identify FN & TN ✓ Correct

FN = 10, TN = 130

Diabetics missed by the model = 50 − 40 = 10 → FN = 10. Healthy patients correctly cleared = 150 − 20 = 130 → TN = 130.

Q3 CALC Accuracy ✓ Correct

85%

Accuracy = (TP + TN) / Total = (40 + 130) / 200 = 170 / 200 = 0.85 → 85%.

Q4 CALC Precision ✓ Correct

0.67

Precision = TP / (TP + FP) = 40 / 60 ≈ 0.67. Of all patients the model labelled diabetic, only 67% truly were.

Q5 CALC Recall ✓ Correct

0.80

Recall = TP / (TP + FN) = 40 / 50 = 0.80. The model successfully detected 80% of all actual diabetics.

Q6 CALC F1 Score ✗ Wrong

Correct answer: 0.73

F1 = 2 × (P × R) / (P + R) = 2 × (0.67 × 0.80) / (0.67 + 0.80) = 1.072 / 1.47 ≈ 0.73.

Q7 TRICKY High accuracy ≠ good model ✓ Correct

98%

TN = 980, FN = 20, TP = 0, FP = 0. Accuracy = 980 / 1000 = 98%. Yet it catches zero fraud — a perfect example of why accuracy alone is deceptive on imbalanced data.

Q8 TRICKY Same accuracy, different tradeoff ✓ Correct

Model B — fewer real threats slip through undetected

Missing a real threat (FN) is far costlier than a false alarm (FP) here. Model B's Recall of 0.95 means only 5% of real threats slip through. Model A catches just 60%. Higher Recall is the right priority.

Q9 INTUITION Choosing the right metric ✓ Correct

Recall

Recall = TP / (TP + FN). Maximising Recall minimises False Negatives — the missed cancer cases. In life-critical screening, a false alarm is far preferable to a missed diagnosis.

Q10 INTUITION Choosing the right metric ✓ Correct

Precision

Precision = TP / (TP + FP). Maximising Precision eliminates false alarms — every 'Breaking News' tag must genuinely be breaking news. Incorrectly labelled stories (FP) destroy reader trust.

Q11 SCENARIO Build the full matrix ✓ Correct

TP=160, FP=30, FN=40, TN=270

TP=160 (spam correctly caught). FP=30 (legit wrongly flagged). FN=200−160=40 (spam that slipped through). TN=300−30=270 (legit correctly cleared). Verify: 160+30+40+270=500 ✓

Q12 CALC Precision & Recall from the matrix ✓ Correct

Precision=0.84, Recall=0.80

Precision = 160/(160+30) = 160/190 ≈ 0.84. Recall = 160/(160+40) = 160/200 = 0.80.

Q13 TRICKY Flipping all predictions ✗ Wrong

Correct answer: 50

Reversing predictions swaps TP ↔ TN and FP ↔ FN. The original TN = 50 becomes the new TP = 50.

Q14 INTUITION F1 vs Accuracy on imbalanced data ✓ Correct

Model Y — high F1 reflects balanced precision and recall

Model X's high accuracy paired with a low F1 of 0.45 is a textbook sign of class imbalance — it's probably predicting the majority class almost exclusively. Model Y's F1 of 0.82 shows it handles both classes well.

Q15 TRICKY Perfect Precision, near-zero Recall ✓ Correct

Precision=1.0, Recall=0.013

TP=2, FP=0, FN=148. Precision=2/2=1.0 (never wrong when it flags). Recall=2/150≈0.013 (catches barely 1% of spam). Perfect Precision means nothing if Recall is near zero.

Q16 INTUITION What does a FN really cost? ✓ Correct

False Negatives — real pedestrians the system failed to flag

The pedestrians are the actual positives. The model predicted 'no pedestrian' for all 3 — failing to detect them. That is a False Negative: a real positive the model missed.

Q17 INTUITION Precision–Recall tradeoff in production ✓ Correct

Recall will increase but Precision will likely drop

Lowering the threshold makes the model flag more items. It will catch more real defects (higher Recall / fewer FN), but it will also flag more good products incorrectly (higher FP → lower Precision). This is the classic Precision–Recall tradeoff.